Question	Answer	Marks	Partial Marks
3(a)(iii)	61.9 or 61.92 to 61.93	**3**	**M2** for tan = $\frac{2 \times their\,5 + 5}{their\,5 + 3}$ oe or **B1FT** for 2 × *their* 5 + 5 and *their* 5 + 3
3(b)(i)	28.1 or 28.07 to 28.08	**1**	**FT** *their* 90 – *their* **(a)(iii)** unless *their* **(a)(iii)** < 45, in which case FT *their* **(a)(iii)**
3(b)(ii)	10	**3**	**M2** for $(their\,5 + 3) \times \sqrt{\frac{93.75}{60}}$ oe or **M1** for $\sqrt{\frac{93.75}{60}}$ or $\sqrt{\frac{60}{93.75}}$ oe seen or $\left(\frac{their\,5 + 3}{x}\right)^2 = \frac{60}{93.75}$ oe
4(a)(i)	$1.65 < h \leq 1.8$	**1**	
4(a)(ii)	1.63875	**4**	**M1** for midpoints **soi** **M1** for use of $\sum fh$ with h in correct interval including both boundaries **M1dep on 2nd M1** for $\sum fh \div 80$
4(b)(i)	$\frac{1}{40}$ oe	**1**	
4(b)(ii)	$\frac{63}{395}$ oe	**3**	**M2** for $\frac{56}{80} \times \frac{9}{79}$ [×2] oe or **B1** for $\frac{56}{80}$ or $\frac{9}{79}$ or $\frac{9}{80}$ or $\frac{56}{79}$ oe seen If 0 or B1 scored, instead award **SC2** for answer $\frac{117}{632}$ oe or **SC1** for answer $\frac{63}{400}$ oe
4(c)(i)	15, 39, 71, 80	**2**	**B1** for 3 correct or **M1** for 1 error in addition with other values then consistent

Question	Answer	Marks	Partial Marks
4(c)(ii)	Correct curve	**3**	**B1** for correct horizontal placement for 5 plots **B1FT** for correct vertical placement for 5 plots **B1FT dep on at least B1** for reasonable increasing curve or polygon through *their* 5 points If 0 scored **SC1 FT** for 4 out of 5 points correctly plotted
4(d)(i)	Strict FT *their* UQ – *their* LQ	**2dep**	**B1dep** for *their* UQ or *their* LQ seen Dep on increasing curve/polygon for 2 marks or B1
4(d)(ii)	Strict FT *their* reading at 48	**2dep**	**B1** for 48 written
5(a)(i)	251 or 251.3 to 251.4	**2**	**M1** for $\frac{1}{3}\times\pi\times4^2\times15$ oe
5(a)(ii)	79.5 or 79.51…	**5**	**M3** for $\pi\times4\times\sqrt{4^2+15^2}$ oe or **M2** for $\sqrt{15^2+4^2}$ oe or **M1** for [l^2 =] $4^2 + 15^2$ oe or $\pi\times4\times$ *their* l **M1** for $\frac{\textit{their}\text{ curved surface area}}{\textit{their}\text{ curved surface area}+\pi\times4^2}$ [×100] oe
5(b)(i)	13 min 20 sec	**3**	**B2** for 800 or $\frac{40}{3}$ oe seen or **M1** for figs 3 ÷ figs 375 or figs 3 ÷ 22 500
5(b)(ii)	0.472 or 0.4715 to 0.4716…	**3**	**M2** for $\pi\times0.45^2\times h=0.3$ or $\pi\times45^2\times h=300\,000$ oe or **M1** for $\pi\times\text{figs}45^2\times h=\text{figs}3$ oe
6(a)(i)	$\frac{1}{5}$, $\frac{2}{7}$, $\frac{3}{9}$ final answer	**2**	**B1** for 2 correct terms isw or for 0.2 and (0.286 or 0.2857…) and 0.333…
6(a)(ii)	36	**2**	**M1** for $k=\frac{12(2k+3)}{25}$ or better